

Dr Ioannis Valasakis

SOFTWARE & RESEARCH ENGINEER / MEDICAL IMAGING & CLINICAL XR / ML SYSTEMS

Finding signal in the noise.

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■ 01 PROFILE

Software and research engineer who architects deep-learning systems for clinical use and carries them from model design through to deployed, clinician-facing software. I lead the AI/ML work for atrial-fibrillation detection in the Glasgow Electrocardiology group, designing multi-branch neural networks for 12-lead ECG and building the React and Python/FastAPI viewer that puts them in front of clinicians. Earlier I co-founded and led engineering at a medical VR startup (Unity, Meta headsets) and built deep-learning reconstruction for cardiac MR. My systems background is in C, C++, and CUDA. I work across cardiac signal, neuroimaging, and immersive XR, and own problems end to end with little supervision.

■ 02 SELECTED SOFTWARE IMPACT

Clinical AI	Atrial-fibrillation detection, in the clinic Architected the AF detection network and shipped the React and Python/FastAPI viewer that delivers it to clinicians.
Clinical VR	Medical XR & cardiac imaging Therapeutic VR in Unity for Meta headsets with UCL Hospitals, and a 3D-UNet reconstruction pipeline for cardiac MR. Real-time visualisation, DICOM and NIfTI.
GPU compute	neurogale, whole-brain connectivity at scale Open-source Python/CuPy/Dask framework with a DOI-linked release, roughly 800-fold over a 64-core AFNI baseline on the same workload.

■ 03 EXPERIENCE

Apr 2025 – present	Research Software Engineer, AI/ML Lead <i>University of Glasgow Electrocardiology Section, Glasgow Royal Infirmary (Macfarlane group)</i> Glasgow, UK - Lead the AI/ML work for AF detection, setting the experimental direction and owning the models from architecture design through validation with little day-to-day supervision. - Designed a hierarchical SE-ResNet-1D with an auxiliary V1-lead branch, fusing leads II, aVF, and V1 for 3-class AF/Normal/Other prediction; trained on expert-adjudicated MIMIC-IV-ECG and externally validated on PTB-XL under a blinded protocol. - Built the clinical ECG viewer with a React front end and a Python/FastAPI backend, decoupling the ML pipeline from the display so hospital machines carry no ML dependencies. - Engineered reproducible training and evaluation with PyTorch, MLflow, DVC, and CUDA, added Grad-CAM and SHAP explainability, and translate cardiologist feedback into concrete experiments and thresholds.
Jan 2023 – Nov 2025	Chief Technology Officer & Co-Founder <i>Tycho MedLink (medical VR startup)</i> London, UK - Led Unity development of therapeutic VR prototypes for Meta headsets, taking clinically informed concepts through rapid iteration, real-time interaction design, and evaluation. - Ran small clinical studies with UCL Hospitals, built and led a small engineering team, and helped secure roughly £100k of UKRI funding across two rounds (UCL and Cambridge alumni accelerators).
Oct 2021 – present	PhD Researcher, Computational Neuroscience <i>King's College London (viva passed 2026, award pending minor corrections)</i> London / remote - Built an end-to-end neuroimaging ML pipeline for autism heterogeneity: fMRIPrep-derived rs-fMRI, voxel-wise graph features, 3D DenseNet-121, nested cross-validation, Bayesian hyperparameter optimisation, and saliency-driven subgroup analysis. - Released neurogale, GPU-accelerated Python software for whole-brain connectivity metrics, with CuPy/Dask backends, NIfTI I/O, deterministic controls, tests, documentation, and a DOI-linked Zenodo release. - Presented at OHBM Seoul (2024) and Montreal (2023), and gave invited AI and neuroscience talks (UCL AI Society), to mixed technical and scientific audiences.
Oct 2020 – Sep 2021	Senior Research Engineer <i>Prepaire</i> Dubai / remote Built Python/Luigi/AWS genomic-analysis workflows, reducing turnaround by roughly 40% through standardised pipeline stages, containerised infrastructure, and reproducible statistical reporting.

Jan 2019 – Mar 2020	Senior Software Engineer, Pipeline <i>Saddington Baynes</i>	London, UK
	Built artist-facing production-pipeline tooling for high-resolution visual assets using Python, C++/CUDA, PyTorch, Docker, and web interfaces, supporting asset organisation, CNN-assisted annotation, and real-time iteration.	
2018	Software Engineering Intern, Microsoft <i>Cloud-hypervisor and Linux-kernel engineering in C: systems stability, performance, and low-level debugging.</i>	Cambridge, UK
2016 – 2018	Software Engineer (Systems), Kano Computing <i>Kano OS components and CI/build workflows in Python, C++, Qt/GTK, Docker, Jenkins, Ansible; built a filesystem-diff tool that cut OS image verification from roughly four hours to thirty minutes.</i>	London, UK

■ 04 SELECTED PUBLICATIONS, TALKS & SOFTWARE

- Software.** Valasakis, I. *neurogale: GPU-accelerated large-scale exploration for brain connectivity analysis*. Zenodo, 2025. doi:10.5281/zenodo.17419836. github.com/wizofe/neurogale
- Dimitrov, M., Wong, N. M. L., Leaman, S., França, L. G. S., Valasakis, I., et al. *Sensorimotor functional connectivity is differentially modulated in autistic individuals by the μ -opioid agonist tianeptine*. *Molecular Psychiatry* (in review), 2025.
- Valasakis, I., Holiga, Š., Deprez, M., Batalle, D. *Subtyping autism based on non-linear features extracted from fMRI and explainable DenseNet architectures*. OHBM Annual Meeting, Seoul, 2024. Invited talk on this work, UCL AI Society, 2024.
- Valasakis, I., Holiga, Š., Deprez, M., Batalle, D. *Towards autism subtypes using fMRI, network connectivity metrics, and deep-learning explainable features*. OHBM Annual Meeting, Montreal, 2023.
- Valasakis, I., Qi, H., et al. *Deep-learning reconstruction using a 3D-UNet for cardiac MR angiography imaging*. ISMRM Annual Meeting, 2021.

■ 05 EDUCATION

2021 – present	PhD, Computational Neuroscience <i>King's College London (viva passed 2026, award pending minor corrections)</i> <i>Thesis: Mapping the Spectrum: Voxelwise Functional Connectivity, Interpretable Deep Learning, and Data-Driven Subtyping of Autism Heterogeneity</i>	
2020	MRes, Healthcare Technologies (Distinction) <i>King's College London: biomedical imaging and deep-learning reconstruction.</i>	
2019	MSc, Bioinformatics with Systems Biology <i>Birkbeck, University of London</i>	

■ 06 SELECTED PROJECTS, OPEN SOURCE & TEACHING

2023	Google Summer of Code, Emory University <i>PyQt/PyTorch drag-and-drop desktop app for rapid neural-network prototyping.</i>	Remote
2022	Google Summer of Code, INCF / McGill <i>Python/PyTorch API prototype for infant eye-tracking research, mentored by a clinical ophthalmology group.</i>	Remote
2022	Teaching Assistant, Neuromatch Academy <i>Deep learning in neuroscience: PyTorch, neuroimaging software, FSL, and Nipype for international student teams.</i>	Remote

■ 07 TECHNICAL STRENGTHS

Languages	Python, C, C++, C#, CUDA; Linux, Docker, Git, Jira; pytest, GitHub Actions / Jenkins; profiling, performance optimisation, reproducible pipelines.
Graphics & XR	Unity, Meta VR headsets, real-time visualisation, React front ends, GPU/CUDA computing; FastAPI / RESTful APIs; AWS, Azure, SLURM/HPC.
ML & medical	PyTorch, JAX, scikit-learn; 1D/3D CNNs, SE-ResNet, DenseNet, transformers; MLflow, DVC; explainability (Grad-CAM, SHAP); DICOM, Nifti, ECG/WFDB; fMRIPrep, FSL, Nipype; Rust.